

Supplementary Material: Positive-Class Weighting Induces Systematic Probability Distortion in Deep Learning Models for Chest Radiography under Temporal Cohort Shift

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Section S1: Comprehensive Cross-Institutional Factorial Matrix

Table S1 provides the complete $2 \times 2 \times 4$ cross-institutional factorial matrix on the official MIMIC-CXR-JPG test cohort ($N = 3,403$ frontal radiographs from 3,041 diagnostic studies across $N = 289$ unique patients). Both unweighted and positive-weighted binary cross-entropy (BCE) models are evaluated across DenseNet-121 and ResNet-50 architectures for all four thoracic abnormalities. For positive-weighted models, paired zero-parameter analytic logit recalibration ($z \mapsto z - \log w$) is evaluated alongside patient-clustered 1,000-replicate bootstrap 95% confidence intervals.

Section S2: Diagnostic Study-Level Sensitivity Analysis

To evaluate sensitivity to repeated within-patient examinations and multi-image studies, Table S2 summarizes external validation performance aggregated strictly at the diagnostic study level ($N = 3,041$ studies across $N = 289$ unique patients). Multiple images acquired during a single clinical study are aggregated using mean probability pooling. Concordance between Table S1 and Table S2 confirms that all primary conclusions are invariant to whether the statistical unit of analysis is defined per radiograph or per clinical study.

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Table S1: Supplementary Table S1: Complete 16-Row Cross-Institutional External Validation Matrix on the Official MIMIC-CXR-JPG Frontal Test Cohort ($N = 3,403$ Radiographs, 3,041 Studies, 289 Unique Patients). Evaluated on 3-Seed Probability Ensembles with Image Prevalence (π), Empirical Null Reference Baselines ($\text{Brier}_{\text{null}} = \pi(1 - \pi)$), Patient-Clustered Bootstrap 95% Confidence Intervals, Brier Skill Scores ($\text{BSS} = 1 - \text{Brier}/\text{Brier}_{\text{null}}$), and Paired Analytic Recalibration ($\Delta\text{Brier} = \text{Brier}_{\text{corr}} - \text{Brier}_{\text{raw}}$).

Architecture	Objective	Pathology	Prevalence (π)	$\text{Brier}_{\text{null}}$	AUROC [95% CI]	Brier (Raw $\rightarrow$ Corr)	ΔBrier [95% CI]	BSS (Raw $\rightarrow$ Corr)
DenseNet-121	Unweighted BCE	Pleural Effusion	32.18%	0.2182	0.688 [0.656, 0.717]	0.2488	–	-0.140
DenseNet-121	Unweighted BCE	Cardiomegaly	26.33%	0.1940	0.523 [0.491, 0.556]	0.2375	–	-0.224
DenseNet-121	Unweighted BCE	Pneumonia	10.05%	0.0904	0.607 [0.560, 0.650]	0.0986	–	-0.091
DenseNet-121	Unweighted BCE	Edema	21.33%	0.1678	0.562 [0.524, 0.596]	0.2124	–	-0.266
DenseNet-121	Pos-Weighted BCE	Pleural Effusion	32.18%	0.2182	0.668 [0.636, 0.700]	0.2536 $\rightarrow$ 0.2451	-0.0084 [-0.0401, +0.0240]	-0.162 $\rightarrow$ -0.123
DenseNet-121	Pos-Weighted BCE	Cardiomegaly	26.33%	0.1940	0.525 [0.492, 0.558]	0.2907 $\rightarrow$ 0.2328	-0.0579 [-0.0790, -0.0349]	-0.499 $\rightarrow$ -0.200
DenseNet-121	Pos-Weighted BCE	Pneumonia	10.05%	0.0904	0.598 [0.553, 0.639]	0.1078 $\rightarrow$ 0.0983	-0.0095 [-0.0170, -0.0016]	-0.192 $\rightarrow$ -0.087
DenseNet-121	Pos-Weighted BCE	Edema	21.33%	0.1678	0.581 [0.549, 0.611]	0.1914 $\rightarrow$ 0.2131	+0.0217 [+0.0118, +0.0307]	-0.141 $\rightarrow$ -0.270
ResNet-50	Unweighted BCE	Pleural Effusion	32.18%	0.2182	0.677 [0.645, 0.709]	0.2430	–	-0.113
ResNet-50	Unweighted BCE	Cardiomegaly	26.33%	0.1940	0.519 [0.489, 0.548]	0.2333	–	-0.203
ResNet-50	Unweighted BCE	Pneumonia	10.05%	0.0904	0.583 [0.540, 0.623]	0.0982	–	-0.086
ResNet-50	Unweighted BCE	Edema	21.33%	0.1678	0.516 [0.480, 0.552]	0.2126	–	-0.266
ResNet-50	Pos-Weighted BCE	Pleural Effusion	32.18%	0.2182	0.620 [0.588, 0.653]	0.2665 $\rightarrow$ 0.2557	-0.0109 [-0.0417, +0.0253]	-0.221 $\rightarrow$ -0.172
ResNet-50	Pos-Weighted BCE	Cardiomegaly	26.33%	0.1940	0.522 [0.491, 0.552]	0.2822 $\rightarrow$ 0.2211	-0.0611 [-0.0818, -0.0415]	-0.455 $\rightarrow$ -0.140
ResNet-50	Pos-Weighted BCE	Pneumonia	10.05%	0.0904	0.589 [0.545, 0.629]	0.1137 $\rightarrow$ 0.0982	-0.0155 [-0.0254, -0.0059]	-0.257 $\rightarrow$ -0.086
ResNet-50	Pos-Weighted BCE	Edema	21.33%	0.1678	0.570 [0.538, 0.602]	0.1929 $\rightarrow$ 0.2130	+0.0201 [+0.0092, +0.0313]	-0.149 $\rightarrow$ -0.269

Table S2: Supplementary Table S2: Diagnostic Study-Level Sensitivity Analysis on the Official MIMIC-CXR-JPG Test Cohort ($N = 3,041$ Studies across $N = 289$ Unique Patients). Probabilities within multi-image studies are averaged to yield one prediction per study. Evaluated on 3-seed probability ensembles with patient-clustered bootstrap 95% confidence intervals.

Architecture	Objective	Pathology	Study Prevalence	Study $\text{Brier}_{\text{null}}$	AUROC [95% CI]	Brier Score
DenseNet-121	Unweighted BCE	Pleural Effusion	32.55%	0.2196	0.695 [0.663, 0.723]	0.2471
DenseNet-121	Unweighted BCE	Cardiomegaly	26.54%	0.1951	0.526 [0.497, 0.559]	0.2359
DenseNet-121	Unweighted BCE	Pneumonia	10.16%	0.0913	0.608 [0.562, 0.650]	0.0980
DenseNet-121	Unweighted BCE	Edema	21.64%	0.1697	0.565 [0.528, 0.600]	0.2118
ResNet-50	Unweighted BCE	Pleural Effusion	32.55%	0.2196	0.682 [0.650, 0.714]	0.2415
ResNet-50	Unweighted BCE	Cardiomegaly	26.54%	0.1951	0.521 [0.494, 0.548]	0.2321
ResNet-50	Unweighted BCE	Pneumonia	10.16%	0.0913	0.584 [0.543, 0.624]	0.0978
ResNet-50	Unweighted BCE	Edema	21.64%	0.1697	0.516 [0.480, 0.553]	0.2120

Section S3: Reproducibility and Code Execution Guide

The complete codebase is organized to support deterministic replication from raw prediction archives through all tables, figures, and statistical tests:

1. **Environment Setup:** Python 3.10+ with PyTorch, torchvision, scikit-learn, scipy, pandas, numpy, and matplotlib.
2. **Stage 3 Recalibration Execution:**

```
python3 scripts/run_stage3_recalibration.py
```

Computes probability ensembles ($\bar{p}_m = \frac{1}{3} \sum_{s=1}^3 p_{m,s}$) across all 8,302 prospective radiographs (16,604 paired model evaluations) for raw, temperature-scaled, analytic offset ($z - \log w$), fitted offset ($z + a^*$), and Platt-scaled models, running 2,000-replicate patient-clustered bootstrap inference.

3. **Publication Artifact Compilation:**

```
python3 scripts/compile_publication_artifacts.py
```

Generates Tables 1–4 in LaTeX and markdown formats, alongside publication figures (Figures 2–6).

4. **Verification Audit:**

```
python3 scripts/audit_cibm.py
```

Verifies all formatting, word count, highlight length (≤ 85 characters), and statutory compliance rules for Computers in Biology and Medicine.